



COURSE STRUCTURE

Course Code	CSE0PM09A				
Course Category	Program Major				
Course Title	Python Programming				
Teaching Scheme	Lectures	Tutorials	Laboratory / Practical	Project	Total
Weekly load hours	3	—	2	—	5
Credits	3	—	1	—	4
Assessment Schema Code	TL3				

Prerequisites:

First Year pass / ATKT

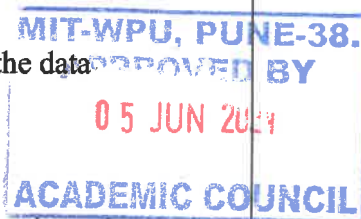
Course Objectives:

1. To introduce programming concepts using python
2. Student should be able to develop Programming logic using python
3. To develop basic concepts and terminology of python programming
4. To test and execute python programs
5. Develop cost-effective robust applications using the latest Python trends and technologies.

Course Outcomes:

After completion of this course students will be able to:

1. Develop a Python program to demonstrate use of operators and data types.
2. Determine the methods to create and develop Python programs by utilizing the data structures like lists, dictionaries, tuples and sets.
3. Develop a python program using Function, modules and packages.
4. Design Classes for a given problem.
5. To understand the application of Python programming using machine learning in various domains, Security, NLP, healthcare.



Course Contents:

Unit 1: Introduction, Data Types and Operators

History of Python, Need of Python Programming, Applications Basics of Python Programming, Features of Python, Python environment setup-Installation and working of IDE, Running Python Scripts, Variables, Assignment, Keywords, Input-Output, Indentation. Python Data Types: Numbers, String, Tuples, Lists, Dictionary and use of data types. Python Control Flow statements and Operators Control Flow Statements: Conditional statements using If statement, if ...else statement, elif statement

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, While loop, For loop, Nested Loop, Break statement, Continue statement, Pass statement, Python's operators: Types of Operators Arithmetic Operators , Comparison Operators , Assignment Operators, Bitwise Operators , Logical Operators, Membership Operators , Identity Operators , Operators Precedence

Unit 2: Data Structures in Python

LISTS: Creating Lists, Basic List Operations, Indexing and Slicing in Lists, Built-In Functions Used on Lists, List Methods,

Tuples: Creating Tuples, Basic Tuple Operations, Indexing and Slicing in Tuples, Built-In Functions Used on Tuples, Relation between Tuples and Lists

Sets: Creating Set, Basic Set Operations, Built-In Functions Used on Set,

Dictionaries: Creating Dictionary, Accessing and Modifying key: value Pairs in Dictionaries, Built-In Functions Used on Dictionaries, Dictionary Methods.

Unit-3 Python Functions, Modules and Packages:

Functions – Defining Functions, Calling Functions, Passing Arguments, Keyword Arguments, Default Arguments,

Variable-length arguments, Scope of the Variables in a Function- Global and Local Variables, Use of Python Built-in Function

Modules: Creating modules, import statements, Importing module, Math module, Random module, namespacing and Scoping.

Packages: Python packages, Introduction to PIP, Installing Packages via PIP, Using Standard (e.g.math,scipy,Numpy, matplotlib, panda,etc) and User defined Packages.

Unit-4 Classes and Objects

Object-Oriented Programming OOP in Python: Classes, 'self-variable, Overriding Methods, Data hiding, Data Abstraction, Inheritance- Multiple, Hierarchical and Multilevel inheritance.

Unit 5: Applications of Python Using Machine Learning

Applications of Python using Machine learning in Healthcare, Security, NLP, Case-study using Healthcare, time series data

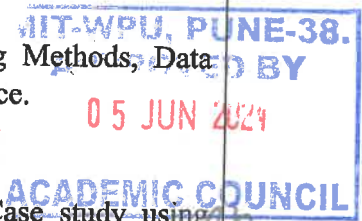
Laboratory Exercises / Practical: Detailed list of experiments. 1 to 12 (Every experiment consist of more than 3 subprograms)

1 Write Python programs for implementing the following.

- Install and configure Python IDE.
- Write a Python Program to display names using Interactive and Script mode.

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- c) Write a Python Program using arithmetic, logical, bitwise and Identity operators.
- d) Write a program to calculate area and perimeter of the square.
- e) Write a program to swap the value of two variables.

2. Write a Python program using conditional statements.

- a) Write a program to check if the input year is leap year or not.
- b) Write a program that takes the marks of 5 subjects and display the grade.
- c) Write a program to check the largest number among the three numbers.

3. Write a Python program using looping statements.

- a) Write a python program to reverse a given number.
- b) Write a python program to print all even numbers between 1 to 100 using while loop.
- c) Print the following pattern.

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4. Write a Python program to perform the following operation on Lists.

- a) Create List, Access List, Update list(Add item, Remove item), Delete List.
- b) Write a Python program to select the even items of a list.
- c) Write a python program to find common items in two lists.

5. Write a Python program to perform the following operation on Tuples.

- a) Create Tuple, Access Tuple, Update Tuple, Delete Tuple.
- b) Write a Python program to find repeated items of a tuple.
- c) Create a tuple and find the minimum and maximum number from it.

6. Write Python program to perform following operation on Dictionary.

- a) Create Dictionary, Access Dictionary, Update Dictionary, Delete Dictionary.
- b) Write a Python program to concatenate dictionaries to create a new one.
- c) Write a Python program to print all unique values in a dictionary.

7. Write a Python program to perform the following operation on Sets.

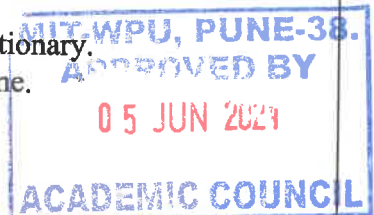
- a) Create Set, Access Set, Update Set, Delete Set
- b) Write a Python program to perform following operations on set: intersection of sets, union of sets, set difference, symmetric difference, clear a set.
- c) Python program to create a set, add member(s) in a set and remove one item from the set.

8. Write a Python program to perform the following operation Function, Modules and Packages.

- a) Write a Python function that accepts a string and calculates the number of uppercase letters and lowercase letters.
- b) Write a Python program that will calculate area and circumference of circle using inbuilt Math module.
- c) Write a NumPy program to generate six random integers between 10 and 30.

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- d) Write a Python program to create user defined module that will ask your college name and will display the name of college.
9. Write a Python program to demonstrate inheritance.
10. Write a program to execute python modules-Pandas,Numpy and ScikitLearn Library(one program for each separately)
11. Case study on healthcare. Consider health care data from UCI/Kaggle.
12. Case study time series data.

Learning Resources:

Reference Books:

1. Rao,K.Naheswara Shaikh Akbar, "Python Programming" Scitech Publications(India) Pvt.Ltd ISBN: 978-9-385-98345-0.
2. Lutz, Mark , "Learning Python" 5th Edition, O'Reilly Publication , Inc. ISBN: 978-1449355739
3. Beazley,David,"Python Essential Reference" 3- Edition,ISBN-13: 978-0-672-32862-6

Supplementary Reading:----

Web Resources:

1. <https://www.w3schools.com/python/>
2. <https://programiz.com/python-programming>
3. <https://www.tutorialspoint.com/python>
4. <https://docs-python.org/3/tutorial/errors.html>
5. <https://spoken-tutorial.org>

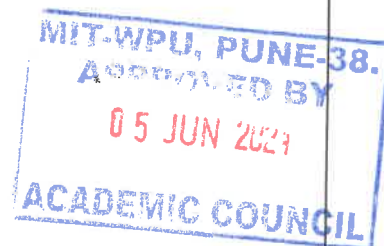
Weblinks:

MOOCs: Online courses for self learning

Courses by NPTEL and MIT Open Courseware etc

Pedagogy:

- Co teaching
- Power point presentations
- Videos
- Demonstrations
- Systematic use of group work and project-based learning.



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Dr. Karad

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